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Byager

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(54) **INSERTER SYSTEM WITH TRANSPORT PROTECTION**

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(57) **ABSTRACT**

Single use inserter system comprising a housing part and a lid part, the inserter system comprising a carrier part in an initial position in the housing and a drive unit comprising a first drive part attached to the housing, the drive unit comprising at least one spring element supported by the first drive part, wherein the drive unit is configured for moving the carrier part from a first position to a second position in relation to the housing in an insertion direction along a first axis. The inserter system comprises a transcutaneous device having a proximal surface and a distal surface, wherein the transcutaneous device is detachably attached to the carrier part, the inserter system comprising at least one transport protection element preventing movement of the carrier part in a direction along the first axis thereby supporting the carrier part in the initial position.

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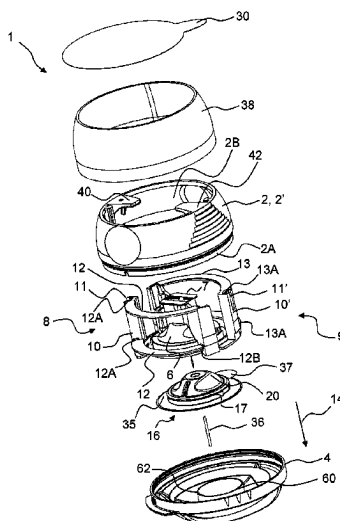
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See application file for complete search history.

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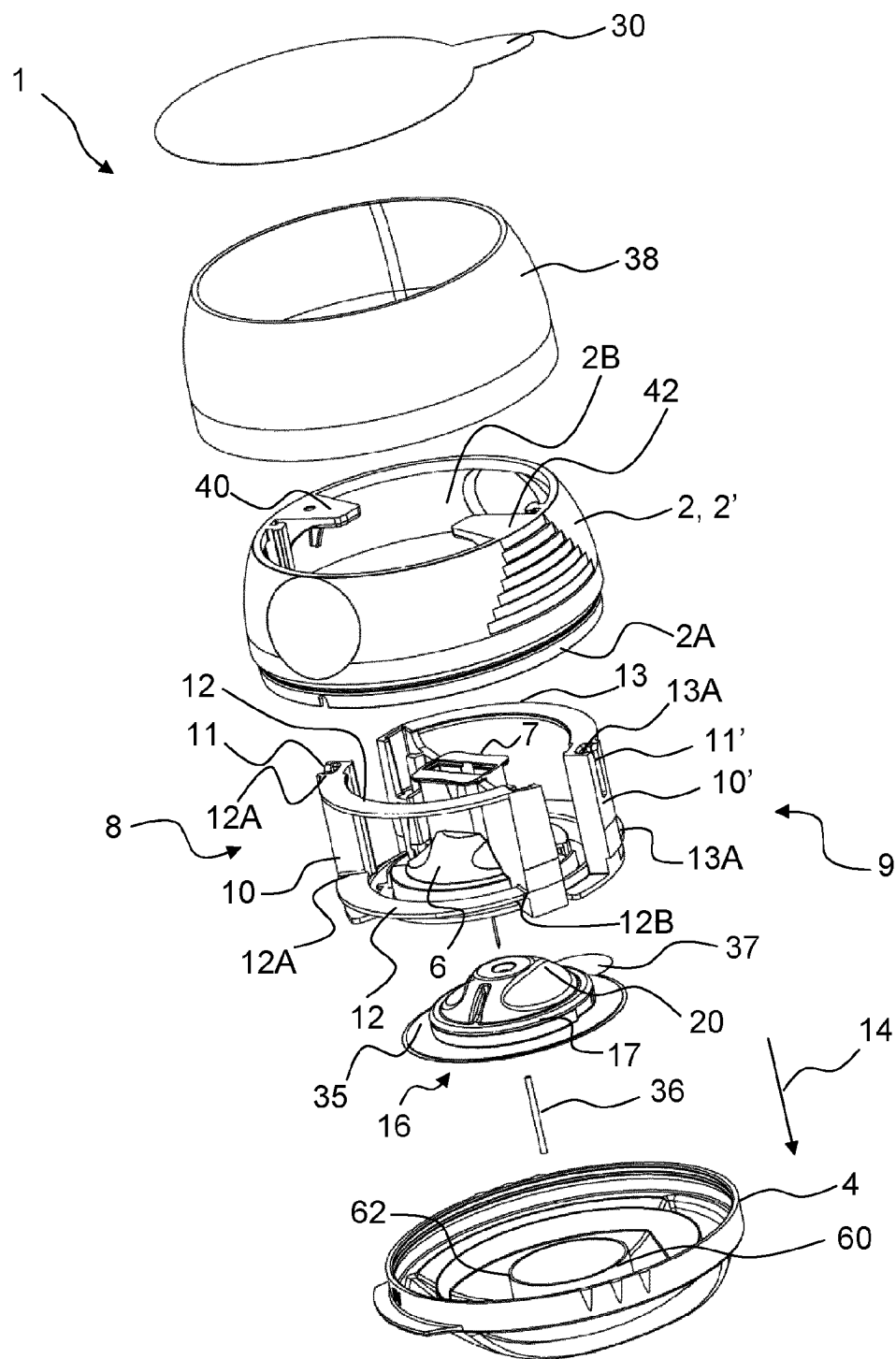


Fig. 1

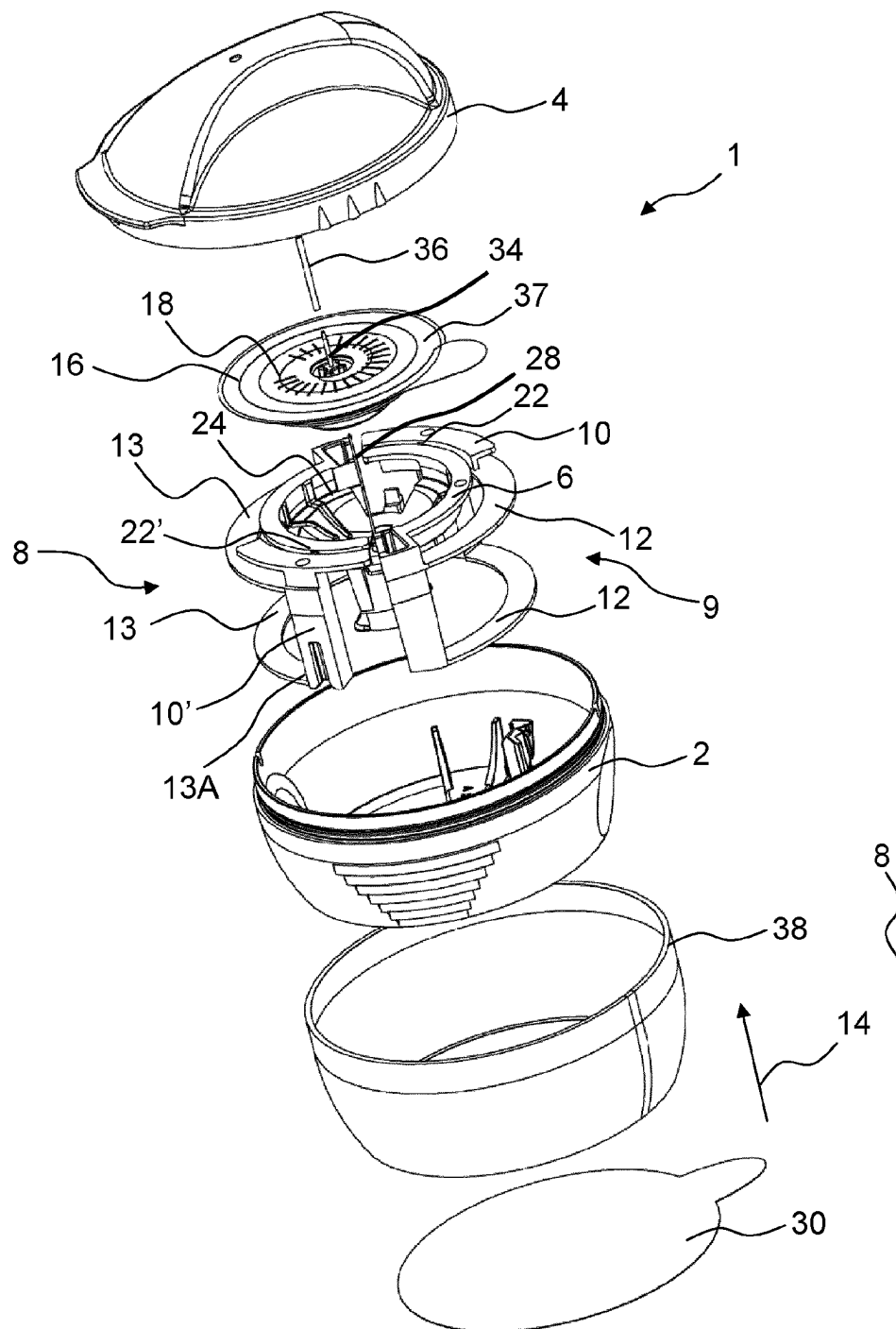


Fig. 2

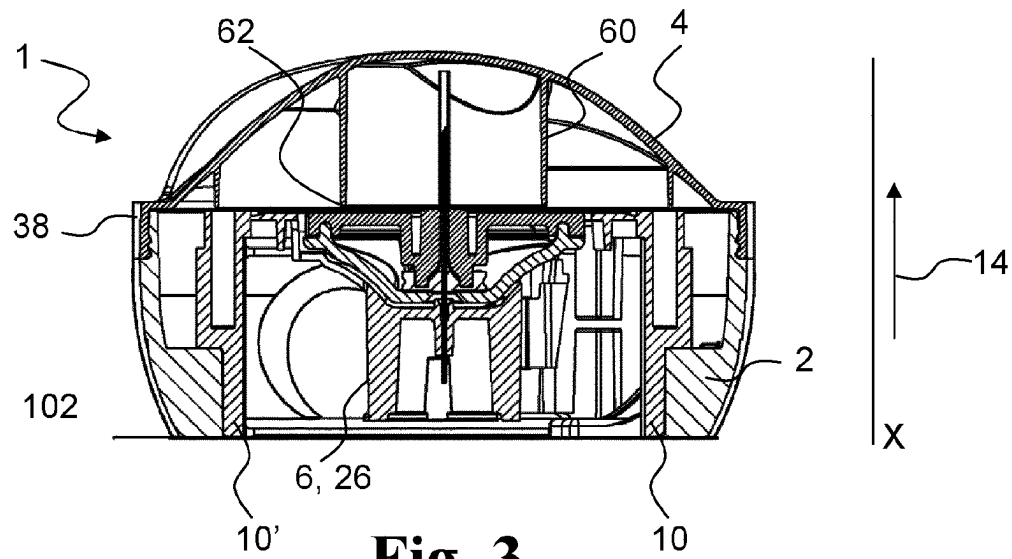


Fig. 3

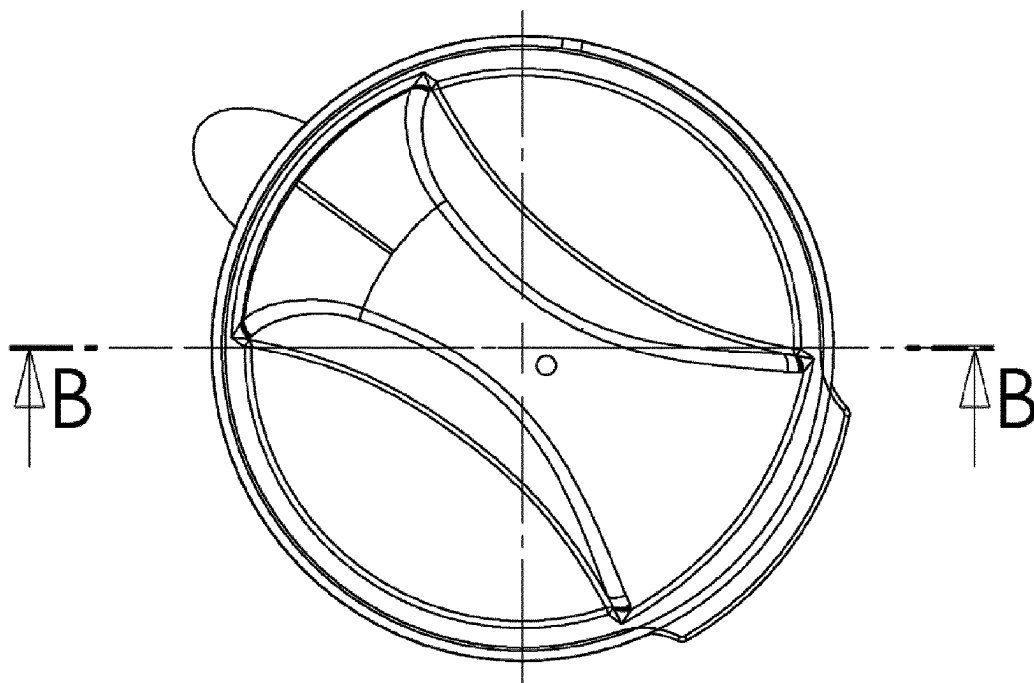


Fig. 4

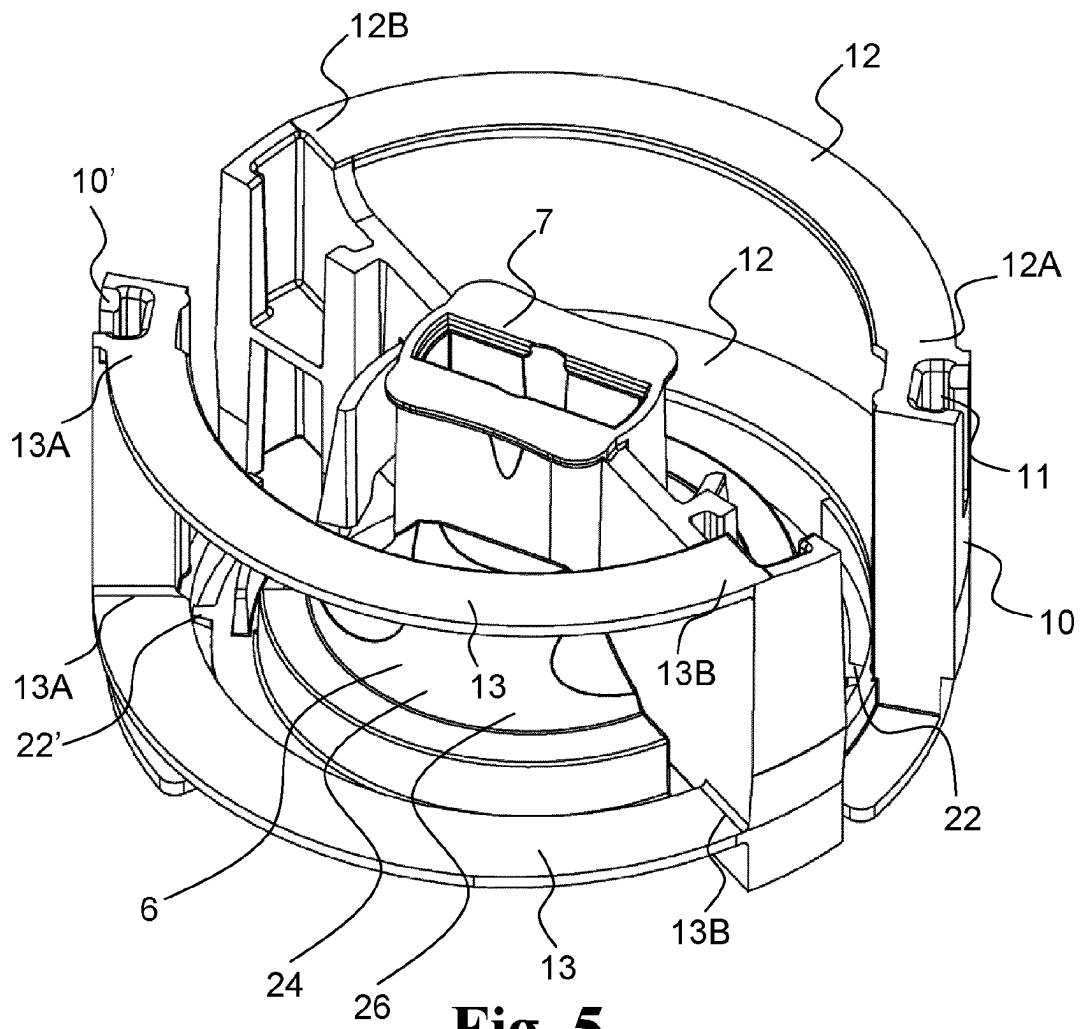


Fig. 5

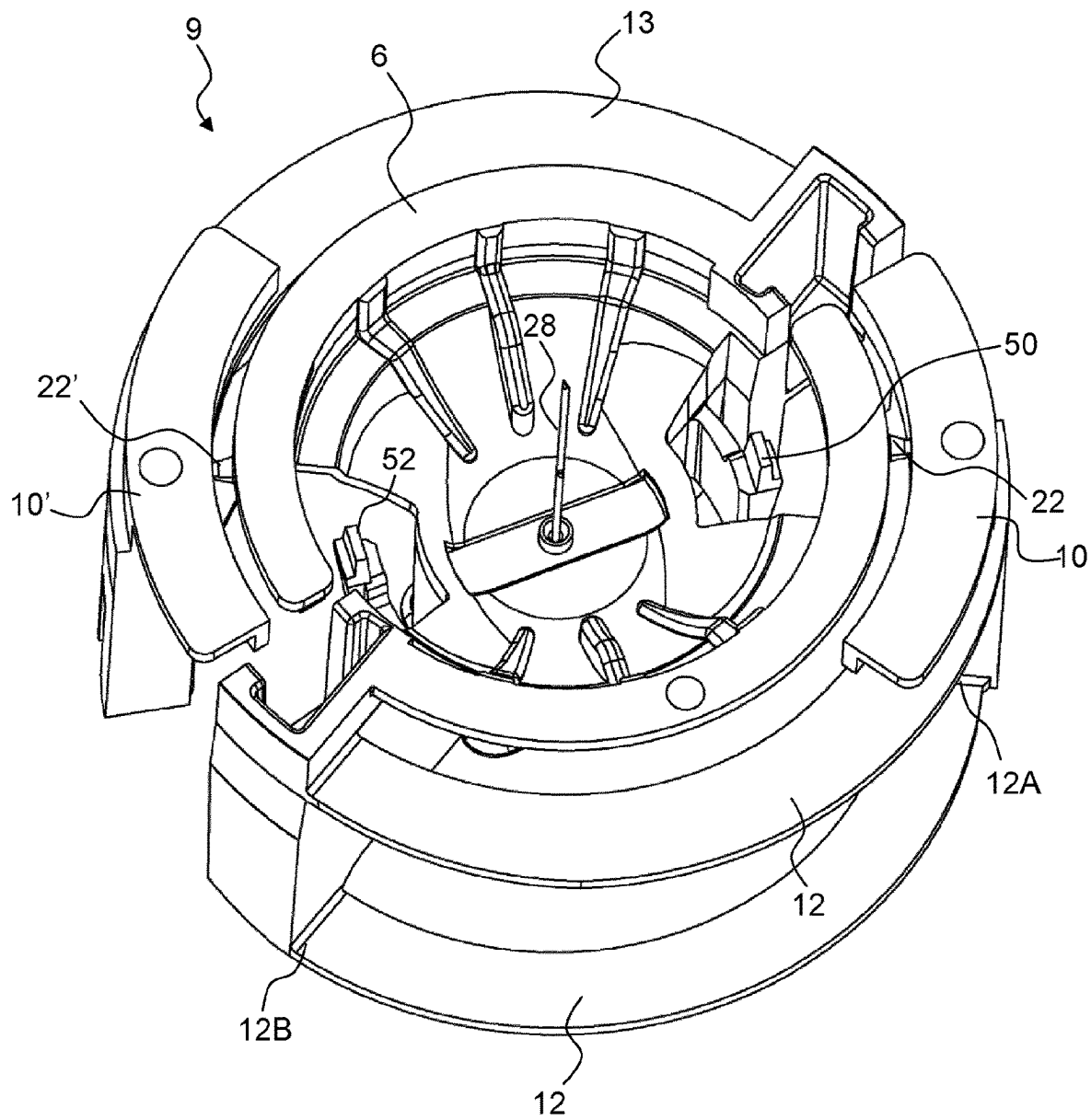


Fig. 6

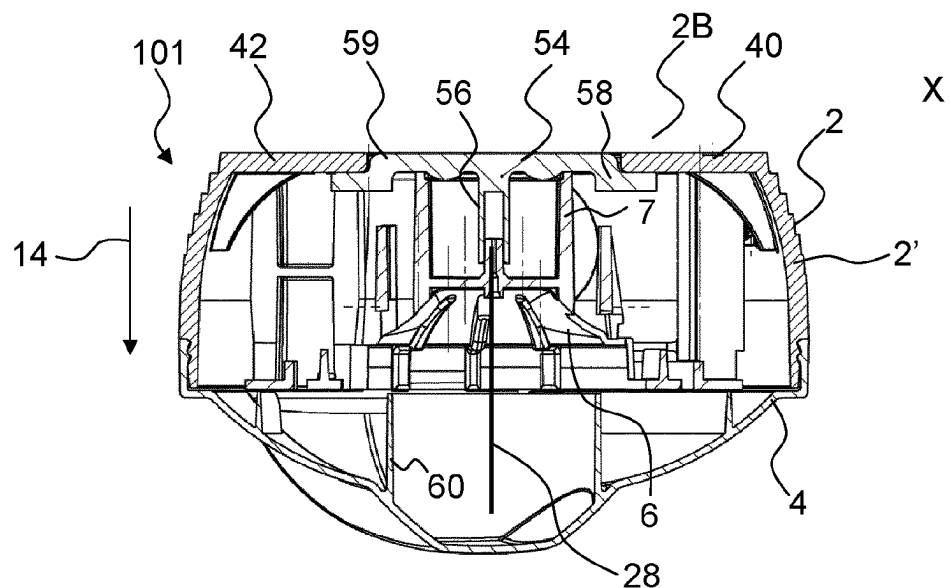


Fig. 7

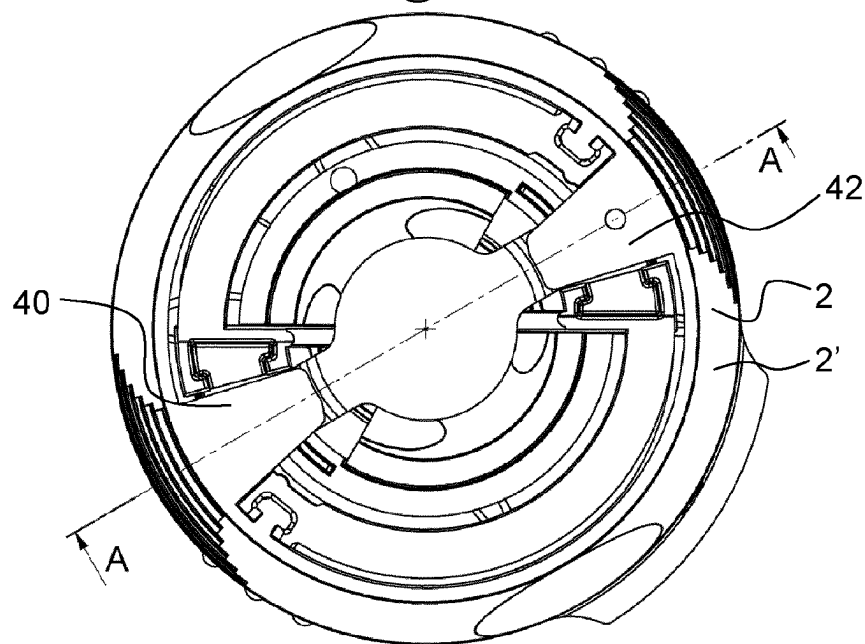


Fig. 8

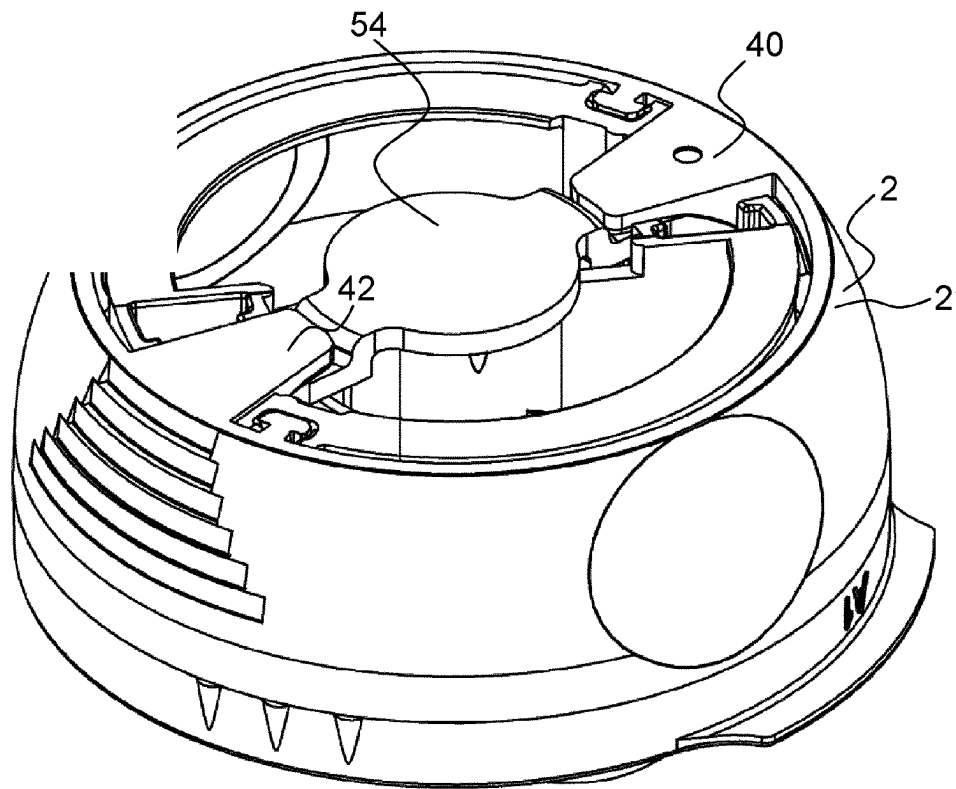


Fig. 9

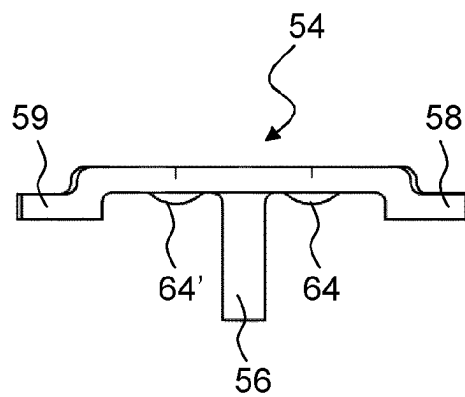


Fig. 10

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INSERTER SYSTEM WITH TRANSPORT PROTECTION

CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 16/265,832, filed Feb. 1, 2019, which is a continuation application of U.S. patent application Ser. No. 14/003,193, filed Nov. 12, 2013, which is a U.S. National Phase of International Application No. PCT/EP2012/053722, filed Mar. 5, 2012, which claims the benefit of U.S. Provisional Application No. 61/452,836, filed Mar. 15, 2011 and EP application Ser. No. 11/158,118.7, filed Mar. 14, 2011, each of which applications is incorporated herein by reference in its entirety and to which applications we claim priority.

BACKGROUND OF THE INVENTION

The present invention relates to an inserter system, in particular a single use inserter system for placing a transcutaneous device on or near the skin of a patient.

Inserter devices that require manual loading of a drive prior to insertion through the skin of a patient typically have a carrier part carrying the transcutaneous device, wherein the carrier part is in an initial position prior to use, i.e. during storage and transportation, wherein the drive is unbiased/unloaded or substantially unbiased. During transportation, the inserter device may be subject to shock or bumps, which may lead to movement of the unbiased carrier part possibly causing breakage of the packaging, e.g. separation of a seal or cover sheet from the housing. Further, shocks or bumps affecting the inserter system may result in displacement of the transcutaneous device in relation to the carrier part. Displacement of the transcutaneous device may lead to discomfort during insertion or even malfunction of the inserter system.

Typically, preloading of inserter devices with spring elements made of a plastic material, such as POM, is not suitable due to the material properties of the spring element material. Accordingly for inserter devices with plastic spring elements, the drive unit is typically unloaded or substantially unloaded in the initial or transport position, thus requiring loading of a drive unit of the inserter device prior to placing or inserting a transcutaneous part of the inserter device.

SUMMARY

There is a need for inserter systems with capability of withstanding shocks and bumps, e.g. during transportation or relocation.

Accordingly, an inserter system is provided, the inserter system being a single use inserter system comprising a housing part and a lid part, the inserter system comprising a carrier part in an initial position in the housing and a drive unit comprising a first drive part attached to the housing, the drive unit comprising at least one spring element supported by the first drive part, wherein the drive unit is configured for moving the carrier part from a first position to a second position in relation to the housing in an insertion direction along a first axis. The inserter system may comprise a transcutaneous device having a proximal surface and a distal surface, wherein the transcutaneous device is detachably attached to the carrier part, and the inserter system may comprise at least one transport protection element preventing movement of the carrier part in a direction along the first axis thereby supporting the carrier part in the initial position.

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The inserter system provides an improved inserter system with increased strength and which is able to withstand shocks and bumps typically experienced during transport and storage without damaging the inserter system.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other features and advantages of the present invention will become readily apparent to those skilled in the art by the following detailed description of exemplary embodiments thereof with reference to the attached drawings, in which:

FIG. 1 is an exploded view of an exemplary inserter system,

FIG. 2 is a different exploded view of the inserter system in FIG. 1,

FIG. 3 shows a cross section of the inserter system in FIG. 1,

FIG. 4 shows the inserter system of FIG. 1 perpendicular to the insertion direction,

FIG. 5 shows a perspective view of a first unit of an inserter system,

FIG. 6 shows another perspective view of the first unit in FIG. 5,

FIG. 7 shows a cross section of an inserter system,

FIG. 8 shows the inserter system of FIG. 7,

FIG. 9 shows a perspective view of the inserter system in FIG. 7, and

FIG. 10 shows a locking member.

The figures are schematic and simplified for clarity, and they merely show details which are essential to the understanding of the invention, while other details have been left out. Throughout, the same reference numerals are used for identical or corresponding parts.

DETAILED DESCRIPTION OF THE INVENTION

The inserter system of the present invention alleviates the problems experienced with inserter devices with unbiased drive units.

The inserter system comprises a housing part and optionally a lid part. The housing part has a sidewall with a first end and a second end. A first opening and/or a second opening may be provided at the first end and second end, respectively. When assembled, the lid part may cover the first opening at the first end.

The inserter system may comprise a gas permeable seal or cover sheet. The seal may cover the second opening at the second end of the housing part. The housing part, the seal, and optionally the lid part may define a closed cavity accommodating the transcutaneous device in a sterile environment prior to use. The gas permeable seal allows sterilization of the inserter system upon assembly.

The inserter system comprises a transcutaneous device comprising a base with a distal surface and a proximal surface, the transcutaneous device comprising a transcutaneous element, such as a cannula and/or a sensor element, mounted to the base extending from the proximal surface facing the skin of a patient. The proximal surface may be an adhesive surface for securing the transcutaneous device to a base plate and/or on the skin of the patient. A mounting pad with an adhesive surface may be attached to the proximal surface of the base of the transcutaneous device. The inserter system may comprise a protective layer covering the adhesive surface. The transcutaneous device may comprise locking elements for locking engagement with a base plate.

Prior to use, the carrier part is in an initial position. Prior to inserting the transcuteaneous device, a user has to load the inserter system. During loading of the inserter system, the carrier part and the transcuteaneous device are moved by the user from the initial position to a first position also referred to as a loaded position or retracted position, e.g. in a retraction direction opposite the insertion direction. By moving the carrier part, e.g. along the first axis and/or around the first axis, to the first position, the drive unit connected to or supporting on the carrier part is biased or loaded for moving the carrier part and the transcuteaneous device from the first position to a second position also referred to as an injection position. In the first position, the carrier part is releasably locked by one or more first locking elements, such as a first primary locking element and/or a first secondary locking element on the carrier part. In the first position, the first locking elements may support on a stationary part of the inserter system including the housing part and/or the first drive part. After loading, the inserter system is positioned on the skin and the carrier part is released. Upon release of the first locking element(s), the loaded drive unit moves the carrier part and the transcuteaneous device in the insertion direction to the second position placing the transcuteaneous element under the skin. After insertion, the carrier part and housing part are removed and the transcuteaneous element, optionally including tubing, is ready for use.

The carrier part of the inserter system may comprise a needle hub with a hub base and optionally an insertion needle secured to the hub base, wherein the transcuteaneous device is detachably attached to the needle hub. The carrier part may comprise a carrier base, wherein the needle hub is releasably attached to the carrier base. The inserter device may be configured for releasing the needle hub from the carrier base, e.g. in the second position. The inserter device may be configured for moving the needle hub to a third position in the extraction direction upon release from the carrier base in the second position.

The carrier part, with or without an insertion needle, and the drive unit may be molded in a single first unit reducing assembly costs.

The inserter system comprises a stationary part. The stationary part is the parts of the inserter system that are stationary in relation to the skin of a patient when the transcuteaneous device is inserted. The stationary part comprises the housing part and at least the first drive part(s).

The at least one transport protection element provides a releasable support or locking of the carrier part and/or transcuteaneous device in the initial position, thereby preventing or limiting movement of the carrier part and/or transcuteaneous device in the insertion direction and/or in the extraction direction opposite the insertion direction along the first axis. Thus, the transport protection elements may provide that the transcuteaneous device in the initial position is substantially fixed or secured in relation to the carrier part thereby avoiding user discomfort or malfunction of the inserter system during use. Further, the transport protection element(s) may provide fixation of the carrier part and/or the transcuteaneous device in relation to the stationary part reducing or eliminating the risk of breaking the seal or cover sheet during transportation.

The at least one transport protection element may comprise at least one bridge between the carrier part and the stationary part of the inserter system. The at least one bridge has dimensions or weakened sections that enable a user to break the bridges when loading the inserter system in order to move the carrier part to the first position in relation to the

stationary part. At the same time, the bridge dimensions are selected to withstand forces resulting from transport bumps or shocks, thereby securing the carrier part in the initial position.

A bridge may comprise one or more weakened sections. A weakened section may have a minimum width in the range from about 0.1 mm to about 2.0 mm, such as in the range from 0.3 mm to 1.5 mm, e.g. about 1.0 mm. A weakened section may have a minimum thickness in the range from about 0.1 mm to about 2.0 mm, such as in the range from 0.3 mm to 1.5 mm, e.g. about 0.5 mm. A weakened section of a bridge may have a minimum cross sectional area in the range from 0.1 mm² to about 4.0 mm², such as in the range from about 0.3 mm² to about 2.0 mm², e.g. 0.5 mm², 0.8 mm² or 1.0 mm².

The first unit may be made of a suitable polymer material such as polyoxymethylene (POM).

The at least one transport protection element may comprise a locking member detachably mounted in the housing. The locking member may engage with or supporting on the carrier part and/or a stationary part of the inserter system, such that movement of the carrier part in a direction, e.g. the insertion direction and/or the extraction direction, along the first axis in relation to the stationary part is limited or substantially prevented, thereby supporting the carrier part in the initial position. The locking member may limit or substantially prevent movement of the transcuteaneous device in the extraction direction and/or in the insertion direction.

The at least one transport protection element may comprise at least one support element formed in the lid part. The at least one support element may include a first support element for supporting the carrier part and the transcuteaneous device in the initial position. The first support element may have a first end positioned adjacent to the proximal surface of the transcuteaneous device limiting or substantially preventing movement of the transcuteaneous device in the insertion direction.

A first support element may be formed as a cylindrical tube having a first end and a suitable cross section, such as circular with a first diameter d_1 , rectangular, oval, or any other suitable shape. The first support element may extend perpendicular to the first axis. The first support element may be arranged to support the transcuteaneous device in the initial position by the first end of the first support element forming a stop member in the insertion direction. The distance between the first end of the first support element and the proximal surface of the transcuteaneous device should be small enough to prevent displacement of the transcuteaneous device in relation to the carrier part and/or the stationary part in the insertion direction prior to use, e.g. during storage or transportation. The distance between the first end of the first support element and the proximal surface of the transcuteaneous device may be less than 2 mm, such as less than 1 mm.

The at least one support element may comprise one or more rods or plate structures, each having an end or edge adjacent to or contacting the transcuteaneous device for preventing displacement of the transcuteaneous device in relation to the carrier part and/or the stationary part in the insertion direction prior to use.

In the initial position of the carrier part, the drive unit may be unbiased FIGS. 1 and 2 show exploded views of an exemplary inserter system of the present invention and FIG. 3 illustrates a cross section of the inserter system 1 along the line B in FIG. 4. The inserter system 1 comprises a housing part 2 having a sidewall 2' with a first opening 2A at a first

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end and a second opening 2B at a second end. The inserter system 1 optionally comprises a lid part 4 covering the first opening 2A. Further, the inserter system comprises a carrier part 6 in an initial position in the housing 4 and a drive unit 8 comprising a first drive part attached to the housing. The carrier part 6 and the drive unit 8 form a first unit 9. The carrier part 6 comprises a handle portion 7 for enabling a user to manually load the inserter system by moving the carrier part in the retraction direction opposite the insertion direction 14. The first drive part comprises a first primary drive part 10 and a first secondary drive part 10'. Each first drive part 10, 10' comprises assembly means including an assembly recess 11, 11', respectively, for engaging with corresponding assembly means formed as assembly protrusions in the housing part 2 for mounting the first unit 9 in the housing part 2. The drive unit 8 comprises at least one spring element including two first spring elements in the form of leaf spring elements 12 which at their first ends 12A are connected to the first primary drive part 10 and at their second ends 12B are connected to the carrier part 6. Further, the drive unit 8 comprises two second spring elements in the form of leaf spring elements 13 which at their first ends 13A are connected to the first secondary drive part 10' and at their second ends 13B are connected to the carrier part 6. A configuration with a single spring element such as a helical spring element optionally in combination with guiding means for the carrier part is contemplated. The drive unit 8 is configured for moving the carrier part 6 from a first position to a second position in relation to the housing in an insertion direction 14 along the first axis. The inserter system 1 comprises a transcutaneous device 16 having a base 17 with a proximal surface 18 and a distal surface 20, wherein the transcutaneous device 16 is detachably attached to the carrier part 6 comprising a needle hub 24 including a hub base 26 and optionally an insertion needle 28. The transcutaneous device 16 comprises a transcutaneous element, such as a cannula 34 and/or a sensor element, mounted to the base 17 extending from the proximal surface facing the skin of a patient. The transcutaneous device may comprise a mounting pad 35 secured to the base 17 and having an adhesive proximal surface for securing the transcutaneous device to a base plate and/or on the skin of the patient. A removable protective layer or sheet 37 covers the adhesive surface.

The inserter system 1 comprises at least one transport protection element in the form of a first bridge 22 connecting the first primary drive part 10 and the carrier part 6 in the initial position. Further, the inserter system 1 comprises a second bridge 22' connecting the first secondary drive part 10' and the carrier part 6 in the initial position. A user loading the inserter system breaks the transport protection elements, i.e. bridges 22, 22', when moving the carrier part from the initial position to the first position in relation to the housing. The bridges 22, 22' are molded as a part of the single unit 9 made of POM. The bridges 22, 22' each comprise a weakened section having a minimum width of about 1.0 mm and a minimum thickness of about 0.5 mm for having desired strength to withstand transport shocks and at the same time enabling a user to break the bridges when loading the inserter system.

The housing part 2 comprises a first projecting element 40 and a second projecting element 42 extending from the sidewall 2' and forming a support surface for engagement with locking members 50, 52 of the carrier part 6 for releasably locking the carrier part 6 with the transcutaneous device 16 in the first position.

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The at least one transport protection element comprises at least one support element including a first support element 60 formed in the lid part 4 of the inserter system 1. The first support element 60 is formed as a cylindrical tube having a first end 62 and a circular cross section with a first diameter d_1 , the cylindrical tube extending perpendicular to the first axis. The first support element 60 is arranged to support the transcutaneous device 16 in the initial position by the first end 62 forming a stop member in the insertion direction. The distance between the first end 62 of the first support element 60 and the protective sheet 37 of the transcutaneous device should be small enough, such as less than 2.0 mm, to prevent displacement of the transcutaneous device 16 in relation to the carrier part 6 in the insertion direction prior to use, e.g. during storage or transportation. In one or more embodiments, the first end 62 contacts the protective sheet 37. The at least one support element may comprise one or more rods or plate structures, each having an end or edge adjacent to or contacting the transcutaneous device.

The inserter system 1 optionally comprises a packing film 38 that is shrink-fitted around the housing part and the lid part 4. The packing film 38 is used for showing if the inserter system has been tampered with and therefore may not be sterile.

FIGS. 5 and 6 show different perspective views of the first unit 9 comprising the carrier part 6 and the drive unit 8. The carrier part 6 comprises a first primary locking element 50 and a second locking element 52 for releasably locking the carrier part 6 in the first position by supporting on the projecting elements 40, 42. Alternatively, or in combination, first locking members may be configured to support on first drive part(s) of the drive unit for releasably locking the carrier part in the first position. The locking elements are released by manual deformation of the housing part 2.

FIGS. 7-9 show an inserter system 101 without a transcutaneous device. The inserter system 101 comprises a locking member 54 detachably mounted in the housing part 2. The locking member 54 comprises a first part 56 extending along the first axis X and in the initial position engaging with the handle portion 7 of the carrier part 6. The handle portion 7 forms a cavity, e.g. with rectangular cross section perpendicular to the first axis, for accommodating and supporting the first part 56. Further, the locking member 54 comprises at least one second locking element including a second primary locking element 58 and a second secondary locking element 59 extending perpendicularly to the first axis and supporting on the stationary part (projecting elements 40, 42) of the inserter system 101. Thereby, the carrier part 6 is prevented from moving in the retraction direction in the initial position such that the cover sheet 30 is not damaged by the carrier part moving due to bumps and shocks during transport. In one or more embodiments, the second locking element(s) 58, 59 may be arranged for preventing the carrier part from moving in the insertion direction in the initial position, e.g. by extending into one or more recesses in the stationary part. The locking member 54 may comprise a third locking element (not shown) configured for engaging and locking the carrier part to the locking member 54. The locking member 54 is released by manual rotation, e.g. 90°, of the locking member around the first axis.

FIG. 10 is a cross sectional view of the locking member 54. The locking member 54 comprises at least one holding element 64, 64' for supporting the locking member in the initial position by extending into the cavity of the handle portion 7 preventing unintentional rotation of the locking member 54 when the inserter system is assembled. When the

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locking member **54** is rotated by a user, the carrier part **6** is forced slightly in the insertion direction by the holding elements **64**, **64'**. The locking member **54** is prevented from rotating in the locked initial position, when the lid part **4** with the first support element **60** is assembled with the housing part **2**.

LIST OF REFERENCES

1, **101** Inserter system
 2 Housing part
 2' Sidewall
 2A First opening
 2B Second opening
 4 Lid part
 6 Carrier par
 7 Handle portion
 8 Drive unit
 9 First unit
 10 First primary drive part
 10' First secondary drive part
 12 First spring element
 12A First end
 12B Second end
 13 Second spring element
 13A First end
 13B Second end
 14 Insertion direction
 16 Transcutaneous device
 17 Base
 18 Proximal surface
 20 Distal surface
 22 First bridge
 22' Second bridge
 24 Needle hub
 26 Hub base
 28 Insertion needle
 30 Seal/cover sheet
 34 Cannula
 35 Mounting pad
 36 Needle protector
 37 Protective sheet
 38 Packing film
 40 First projecting element
 42 Second projecting element
 50 First primary locking element
 52 First secondary locking element
 54 Locking member
 56 First part
 58 Second primary locking element
 59 Second secondary locking element
 60 First support element
 62 First end
 64, 64' Holding element
 X First axis
 The invention claimed is:
 1. An inserter system comprising:
 a housing;
 a carrier movably mounted in the housing, wherein the carrier is movable relative to the housing in a first axial direction from an initial position to a retracted position,

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and in a second axial direction from the retracted position to an insertion position, the first axial direction opposite the second axial direction;

a driver mounted in the housing, wherein the driver is operable to move the carrier from the retracted position to the insertion position;

a transcutaneous device attached to the carrier; and

a breakable bridge engaged between the carrier and a portion of the driver when the carrier is in the initial position to thereby support the carrier in the initial position, wherein the breakable bridge is configured to break in response to movement of the carrier from the initial position to the retracted position.

2. The inserter system of claim 1, further comprising at least one locking member configured to selectively limit movement of the transcutaneous device.

3. The inserter system of claim 2, wherein the at least one locking member comprises a first locking member and a second locking member operable to engage the first locking member;

wherein the housing comprises the first locking member; and

wherein the carrier comprises the second locking member.

4. The inserter system of claim 1, further comprising a gas permeable seal or cover sheet.

5. The inserter system of claim 1, wherein the transcutaneous device is detachably attached to the carrier part.

6. An inserter system comprising:

a housing having an open proximal end;

a carrier movably mounted in the housing, wherein the carrier is movable from an initial position to a retracted position, and from the retracted position to an insertion position;

a transcutaneous device attached to the carrier, wherein movement of the carrier from the retracted position to the insertion position moves a portion of the transcutaneous device beyond the proximal end of the housing;

a gas permeable seal or cover sheet covering the open proximal end of the housing; and

a locking member configured to selectively retain the carrier in the initial position to thereby discourage damaging of the gas permeable seal or cover sheet by the carrier and/or the transcutaneous device.

7. The inserter system of claim 6, further comprising a driver mounted in the housing, wherein the driver is operable to move the carrier from the retracted position to the insertion position.

8. The inserter system of claim 7, further comprising a breakable bridge positioned between the carrier and a portion of the driver when the carrier is in the initial position to thereby support the carrier in the initial position.

9. The inserter system of claim 6, wherein the gas permeable seal or cover sheet allows for sterilization of the inserter system.

10. The inserter system of claim 6, further comprising a support configured to limit movement of the transcutaneous device relative to the carrier.

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